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ABN: 74 643 039 285

Structural Engineering Causation & Scope of Work Report

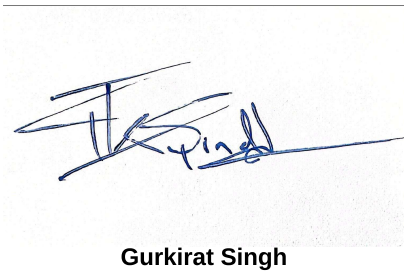
Report reference number:	SE1354
Report date:	22nd of September 2026
Prepared for:	James Goodricke
Site address:	9 Furness Cl Wyndham Vale VIC 3024
Investigation date & time:	27th of August 2026 at 10:00 AM
Prepared by:	Gurkirat Singh

This report is based on a visual inspection limited only to the areas and sections of the property accessible and visible to the inspector on the day of the inspection. The Inspection did not include breaking apart, dismantling, removing, or moving objects including, but not limited to foliage or personal possessions. Please refer to this report's notes and disclaimer section for further information.

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Report Revision Register

Revision	Page No.	Revision Notes	Author	Reviewed by	Date
A	All	Engineering report	Gurkirat Singh	Ali Ibaida	22/09/2026



Gurkirat Singh
Author of Report/Revisions
Signed on behalf of Strength Engineering

Ali Ibaida
Reviewer of Report/Revisions
Signed on behalf of Strength Engineering

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1. Introduction

1.1. Objective:

Strength Engineering Pty Ltd has been engaged as per the brief provided within the **scope of engagement** to undertake an inspection at the **subject property** and report on our findings

1.2. Scope of engagement:

The client has enlisted the services of Strength Engineering to assess and address any structural distress within the property. The primary objectives include identifying the causes behind these issues and formulating a comprehensive plan for rectifying them.

1.3. Investigation methodology:

- Observe and record the extent of damage to the structure as per AS4349.1-2007.
- Provide an opinion as to the cause(s) of the observed damage.
- Report on the findings of the investigation and provide the scope of works recommendations regarding rectification and/or further investigations.

1.4. Documentation referenced:

The documents referenced aid in the formulation of this report.

- AS 2870 2011 Residential Slabs and Footings
- AS 3700 2018 Masonry Structures
- AS 3600 2018 Concrete Structures
- AS 1684 2010 Residential Timber Framed Construction
- CSIRO Guide Managing Foundation Maintenance
- AS/NZS 3500.3:2021 Plumbing and drainage
- Building Regulations 2018
- NCC 2025 National Construction Code Volume Two
- ABCB Housing Provisions Standard 2025

1.5. Subject Property details:

Address:	9 Furness Cl Wyndham Vale VIC 3024
Original Building engineering plans:	Yes
Original Building architectural plans:	No
Building age (Approx):	6 Years Old
Construction type:	Articulated Masonry Veneer
Footing type:	Suspended Waffle Raft
Roofing type:	Tiled Roof
Subject Property contact person:	James Goodricke
Subject Property contact person number:	0431920121
Subject Property contact person email:	jamesandjihyung@gmail.com

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1.6. Inspection details:

Insepction date:	27th of August 2026
Inspecton time:	10:00 AM
Inspection type:	Structural Inspection
Inspection conducted by:	Gurkirat Singh
Weather conditions:	Sunny and Clear Sky

1.7. Limitations:

This report is based on a visual inspection limited only to the areas and sections of the property fully accessible and visible to the inspector on the day of the inspection. The Inspection did not include breaking apart, dismantling, removing, or moving objects including, but not limited to foliage or personal possessions. Please refer to this report's notes and disclaimer section for further information.

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2. Executive Summary

Strength Engineering Pty Ltd conducted an inspection of the subject property as per the scope of engagement outlined by the client.

Key findings include:

- **Floor Level Changes:** Significant differential movements were recorded, with the maximum differential in the dwelling at 54 mm and in the garage at 34 mm, indicating possible foundation instability.
- **Cracking Observations:** Various types of cracking were noted throughout the property, including hairline cracks classified as category 0 and minor stepped cracks classified as category 1, usually attributed to normal settlement and soil moisture variations.
- **Soil Reactivity:** The site is classified as Class 'P', indicating susceptibility to significant ground movement due to moisture changes, necessitating careful footing design considerations.
- **Drainage Issues:** Inadequate surface drainage observed around the property may contribute to increased soil moisture levels, exacerbating foundation movement.
- **Plumbing Conditions:** Potential issues with stormwater and sewer pipes were highlighted, necessitating a full plumbing investigation to ascertain any flaws that may influence foundation stability.
- **Foundation and Structural Integrity:** Concerns regarding the adequacy of foundation construction were raised, particularly with regard to previously performed compliance checks and soil conditions during installation.

Rectification Scope of Works:

1. **Plumbing Investigation:** Engage a suitably qualified plumber to conduct a full investigation of the stormwater and sewer pipes using a CCTV camera system and hydrostatic pressure testing, and to rectify any defects identified in accordance with AS 3500.3 and AS 3500.5.
2. **Drainage Solutions:** Engage a structural engineer to provide a certified design for site surface drainage in accordance with the ABCB Housing Provisions Standard 2025, and a suitably qualified concreter to implement the works, so that all surface water is directed away from the building.
3. **Documentation Review:** Obtain the original geotechnical report, architectural plans, and building permit and inspection records, and provide these to a structural engineer for review to allow the adequacy of the footing system as designed and as constructed to be assessed.
4. **Geotechnical Investigation:** Once the moisture-related issues have been rectified, should the cracking to the garage floor and within the dwelling persist, undertake a geotechnical investigation including boreholes adjacent to the garage to establish the depth and nature of any fill and the founding material.
5. **Localised Repairs:** Following a minimum 6-month dry-off period and once soil moisture conditions have stabilised, engage qualified trades to rectify the observed cracking to the internal linings, external masonry and garage slab, and to reinstate the joint between the external pavement and the brickwork.
6. **Ongoing Monitoring:** Monitor the building following rectification to identify any further movement or development of cracking.

These recommendations aim to address the identified issues comprehensively, ensuring the property's structural integrity and long-term functionality. The summary provides an overview of the report's findings, and a comprehensive review of the entire report is essential for a detailed analysis.

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3. Investigation

3.1. Diagram

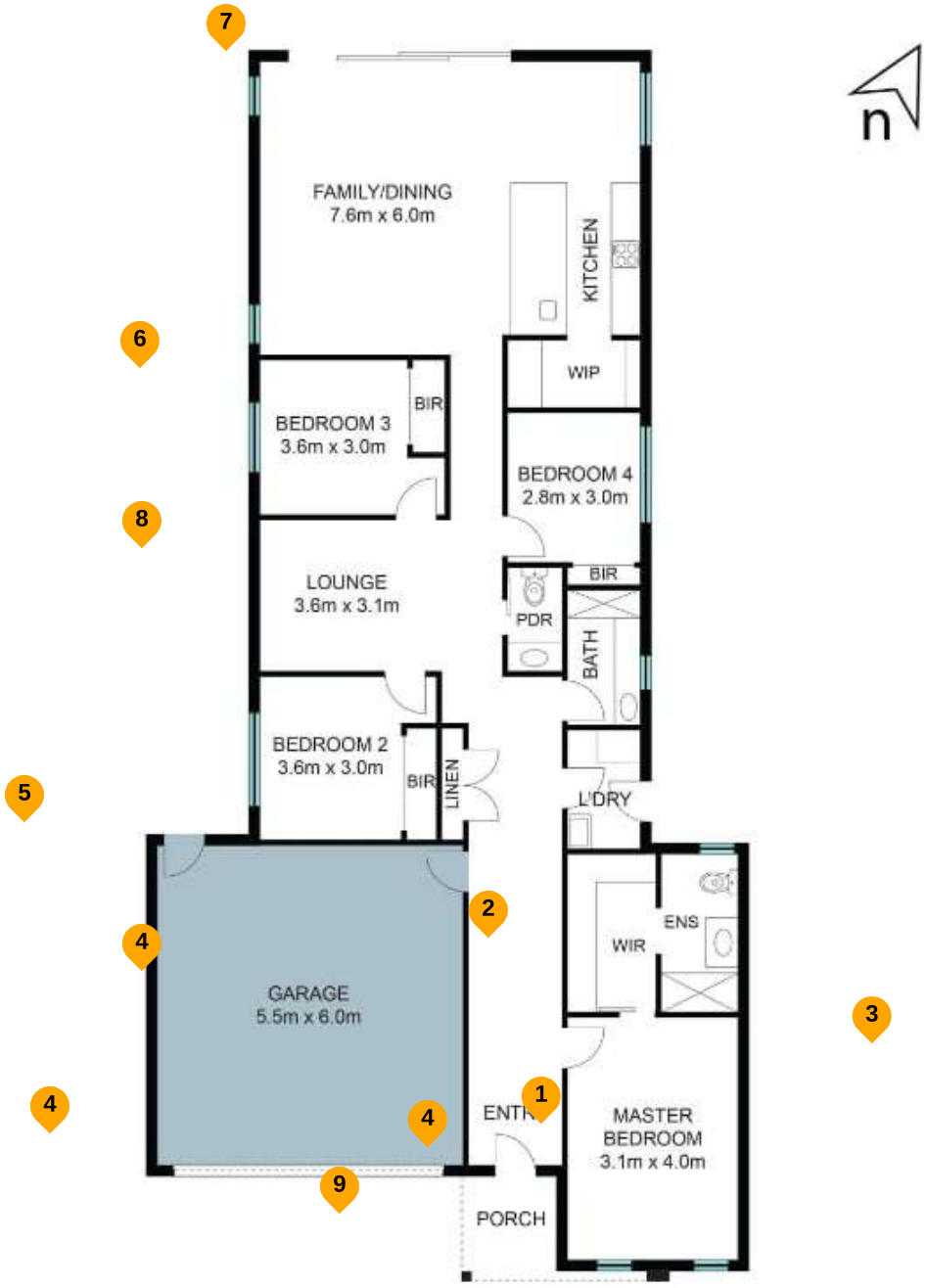


Diagram 1 : The above diagram displays the floor plan of the property and was obtained from realestate.com.au.

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Diagram 2 : Satellite photo of the subject property obtained from Google Maps for illustration purposes

Legend

Property boundary: Blue Line

Areas inspected: Red Line

3.2. Investigation record:

The observations in Table 1 were made during the inspection. Note that this is a general description of the observations that Strength Engineering has deemed relevant to the scope of work. It is not intended to be a complete list of building damage and/or defects.

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3.2.1. Observation 1

Observation No.	1
Photos	
Observation & Comment	A floor level survey was conducted within the footprint of the subject property using a Nivcomp Level NL0001-H25-HS calibrated to +2 mm (Refer to Appendix C). The survey identified significant localised changes in floor

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	level, indicating that sections of the foundation have experienced differing amounts of vertical movement. The dwelling and the garage have been surveyed and assessed as separate floor planes. Within the dwelling, the maximum differential recorded was 54 mm. Within the garage, the maximum differential recorded was 34 mm. The recorded floor levels represent the condition of the dwelling on the day of the investigation only and should not be interpreted as the total amount of historical foundation movement or as an indication that all recorded level differences have resulted from foundation movement, as construction tolerances, previous repairs, renovations, and other factors may also influence the measured floor levels. **Note:** 1. Variations in floor finishes were present at the time of inspection. Although the floor level differences were taken into account, these conditions may have affected the accuracy of the recorded floor level measurements to a degree of ±2 mm. 2. Minor variations in the footings may be due to the footings being constructed slightly unevenly in some areas. The relevant standards allow for construction variations of up to 10 mm.
Causation	The floor level variations are likely due to differential footing movement. Differential settlement occurs when the soil beneath the property compresses, expands or shifts, causing uneven support for the structure and resulting in structural implications such as cracks within the internal and external walls of the subject property. The structure has likely experienced one or more of the following: • **Abnormal soil moisture conditions** – Variations in soil moisture around the building footings can contribute to foundation movement, particularly where reactive soils are present (Class P in this case as per provided document). Abnormal moisture conditions may result from excessive wetting due to plumbing leaks, flooding, inadequate surface drainage or concentrated rainfall runoff, as well as excessive drying associated with trees and other significant vegetation located near the footings. Surface drainage deficiencies were observed at the subject property (Refer to Observation Record 6). • **Site reactivity** – The structural documentation (Refer to Appendix D) classifies the site as Class 'P' in accordance with AS 2870 and nominates an anticipated maximum ground movement of 75 mm. Sites of this reactivity are susceptible to significant seasonal and moisture-related ground movement and require the footing system to be designed and constructed to accommodate that movement.
Scope of Works	**<u>Further investigation</u>** **Plumbing investigation:** • It is recommended that a suitably qualified plumber conduct a full plumbing investigation of the stormwater/sewer pipes, using a CCTV camera system and conducting hydrostatic pressure testing (if possible) to determine if any leakages, blockages, or damages are present. • The plumber is also to comment on the size and specification of the stormwater system and ensure it complies with AS3500.3-2025 Plumbing and drainage standards. • The plumber is to provide a report highlighting the finding to a structural engineer and/or the builder as part of the investigation works. • The plumber is to provide a mud-map outlining the services, including the location and layout of stormwater and sewer pipes, to assist with assessment and rectification works if required.

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- The plumber should also conduct an inspection of the downpipes and gutters to ensure they are appropriately connected and do not present any blockages. Any necessary repairs or replacements should be completed prior to further rectification works.

Scope of works

1. All plumbing defects are to be rectified in accordance with AS 3500.3 and AS 3500.5, ensuring there are no leakages, blockages and/or defects.
2. Surface drainage is to be rectified so that all surface water is directed away from the building (Refer to Observation Record 6).

If plumbing defects allowing water or moisture ingress into the surrounding soil are identified influencing the footing system is removed, a minimum 6-month dry-off period is recommended prior to undertaking footing repairs or crack remediation works to allow the soil moisture conditions to stabilise. During this period, additional footing movement and cracking may occur.

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3.2.2. Observation 2

Observation No.	2
Photos	   
Observation & Comment	The following observation was made near the entrance of the subject property: • Hairline cracking of width less than 0.1 mm, classified damage category 0 (Refer to Table C1 in Appendix A), was observed extending downward from the re-entrant corner of the bulkhead in the corridor of the dwelling. • Further seen minor separation between the cornice-to-wall junction and cornices junction throughout the property. • The observed cracking is classified within damage categories 0 and 1 in accordance with Table C1 of AS 2870-2011 (Refer to Appendix A) and is considered non-structural in nature. • It is important to note that cracking is expected in buildings constructed on normal sites, as outlined in AS 2870–2011 Residential slabs and footings, Clause 1.3. • AS 2870-2011 Residential slabs and footings Cl. B3 states that the major criterion for damage assessment is crack width within walls (Table C1 in Appendix A). For category 1 or 2 damage, remedial action should consist of stabilizing the moisture conditions of the clay (Refer to Observation Record 1).

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1.3.2 Normal sites

Normal sites are those that are classified as one of Classes A, S, M, H1, H2 and E in accordance with Section 2 of this Standard and where foundation moisture variations are those caused by seasonal and regular climatic effects, effect of the building and subdivision, and normal garden conditions without abnormal moisture conditions (see Clause 1.3.3).

NOTES:

1 The application of the recommendations in Appendix B is expected to provide normal garden conditions.

2 Normal sites can be expected to be adversely impacted by irregular climatic effects—this could include prolonged droughts.

1.3 PERFORMANCE OF FOOTING SYSTEMS

1.3.1 General

Buildings supported by footing systems designed and constructed in accordance with this Standard on a normal site (see Clause 1.3.2) that is—

(a) not subject to abnormal moisture conditions; and

(b) maintained such that the original site classification remains valid and abnormal moisture conditions do not develop (see Note 1);

are expected to experience usually no damage, a low incidence of damage category 1 and an occasional incidence of damage category 2 (see Note 2).

Classification of damage shall be as defined in Appendix C.

NOTES:

1 Appendix B provides information and guidance on the maintenance of site foundation conditions.

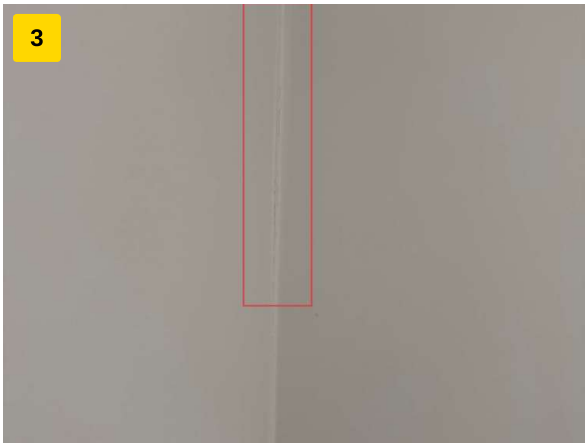
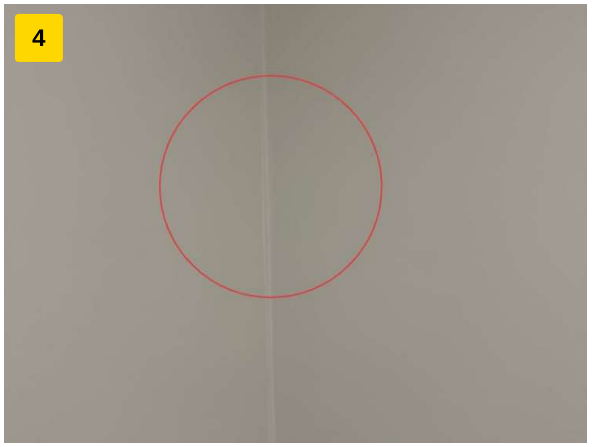
2 Class A sites (as defined in Section 2) are not reactive to moisture and may have a lesser risk of damage to buildings constructed thereon.

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	B3 PERFORMANCE OF WALLS It is acknowledged that minor foundation movements occur on nearly all sites and that it is impracticable to design a footing system that will protect the building from movement under all circumstances. The expected performance of footing systems designed in accordance with the Standard is defined in terms of the damage classifications in Table C1, Appendix C. Crack width is used as the major criterion for damage assessment, although tilting and twisting distortions can also influence the assessment. Local deviations of slope of walls exceeding 1:150 are undesirable. The assessment of damage may also be affected by where it occurs and the function of the building, although these effects are not likely to be significant in conventional buildings. In the classification of damage, account should also be taken of the history of cracking. For most situations Category 0 or 1 should be the limit; however, under adverse conditions, Category 2 should be expected although such damage should be rare. Significant damage is defined as Category 3 or worse. For Category 1 or 2 damage, remedial action should consist of stabilizing the moisture conditions of the clay and paying attention to repairing or disguising the visual damage. This should be regarded as part of the normal maintenance of buildings on reactive clays. Even significant masonry cracking with crack widths over 5 mm often has no influence on the function of the wall and only presents an aesthetic problem. Generally, the remedial action for such damage should start with an investigation to establish the cause of the damage. In many cases the treatment should consist of stabilizing moisture conditions by physical barriers or paths, or replenishing moisture in dry foundations. This may be followed by repair of the masonry and, wherever possible, added articulation should be included while repairs are being effected. Structural repairs to the footing system, such as deep underpinning, should only be considered as the last resort.
Causation	Refer to Observation Record 1
Scope of Works	Strength Engineering recommends the following: • Refer to Observation Record 1 • A qualified painter/plasterer is to be engaged to fill in the crack(s) and gap(s) with a suitable product and repaint the affected area(s) to the desired finish.

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3.2.3. Observation 3

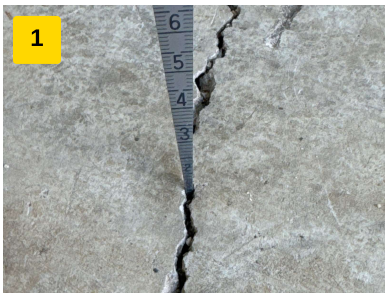
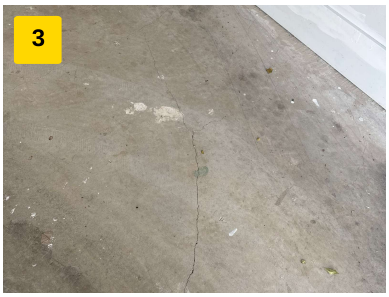
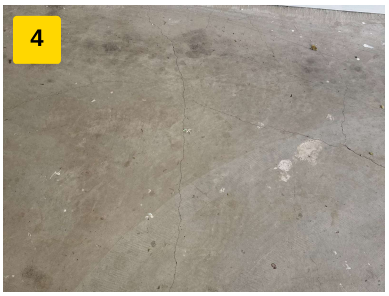
Observation No.	3
Photos	   
Observation & Comment	The following observation was made in the master bedroom of the subject property: • The client advised during the inspection that one of the internal wall corners appeared to be bulging and raised a concern that a structural column at this location may have been affected by the foundation movement. • No cracking was observed to the lining at the corner, and no associated distortion was observed to the adjacent ceiling, cornice, bulkhead or door openings.
Causation	This is likely due to one or more of the following: • External angles in plasterboard lining are typically finished with a metal or plastic corner bead fixed over the arris and set with jointing compound. Where the bead has not been fixed flat against the lining, or where the fixings have subsequently lifted, the bead can sit proud of the surrounding wall surface and present as a raised ridge along the corner. A localised build-up of jointing compound over the bead can produce a similar appearance. • Timber wall framing can bow or twist as it dries following construction, which may cause the lining fixed to it to deviate from plane. Deviation of this nature is typically most pronounced between fixing points, which may explain the protrusion being most apparent at mid-height.

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	Where a load-bearing column or stud pack is present within a wall of this type and has been affected by foundation movement, the resulting distress would ordinarily be accompanied by corroborating signs, including deflection of the beam or bulkhead supported above and cracking of the wall and ceiling linings in the immediate vicinity. No such signs were observed at this location.
Scope of Works	Strength Engineering recommends the following: • Cosmetic repairs may be undertaken, if required, once the moisture-related issues have been addressed and soil moisture conditions have stabilized (Refer to Observation Record 1).

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3.2.4. Observation 4

Observation No.	4
Photos	           
Observation & Comment	The following observations were made in the garage of the subject property: • Several hairline to approximately 2mm wide cracks were observed to the throughout garage concrete floor slab. • A localised dip in the slab surface, with associated discolouration, was observed to the edge of the garage concrete floor slab, as indicated by the red circle and red square on the above images.

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TABLE C2

CLASSIFICATION OF DAMAGE WITH REFERENCE TO CONCRETE FLOORS

Description of typical damage	Approx. crack width limit in floor	Change in offset from a 3 m straightedge centred over defect (see Note 1)	Damage category
Hairline cracks, insignificant movement of slab from level	<0.3 mm	<8 mm	0 Negligible
Fine but noticeable cracks. Slab reasonably level	<1.0 mm	<10 mm	1 Very slight
Distinct cracks. Slab noticeably curved or changed in level	<2.0 mm	<15 mm	2 Slight
Wide cracks. Obvious curvature or change in level	2 mm to 4 mm	15 mm to 25 mm	3 Moderate
Gaps in slab. Disturbing curvature or change in level	4 mm to 10 mm	>25 mm	4 Severe

NOTES:

1 The straightedge is centred over the defect, usually, and supported at its ends by equal height spacers. The change in offset is then measured relative to this straightedge, which is not necessarily horizontal.

2 Local deviation of slope, from the horizontal or vertical, of more than 1:100 will normally be clearly visible. Overall deviations in excess of 1:150 is undesirable.

3 Account should be taken of the past history of damage in order to assess whether it is stable or likely to increase.

Causation

This is likely due to one or more of the following:

- Refer to Observation Record 1
- Due to normal shrinkage of the concrete as it dries and hardens. As the water in the concrete evaporates, the slab slightly contracts, which can cause hairline cracks. The cracks may have been reduced or avoided by providing adequate shrinkage control reinforcement and/or appropriately located saw-cut or construction joints within the garage concrete slab.

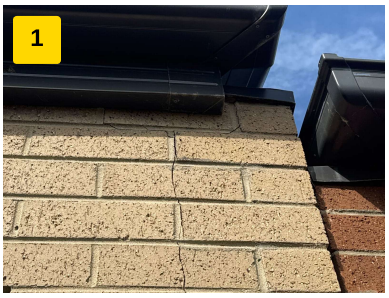
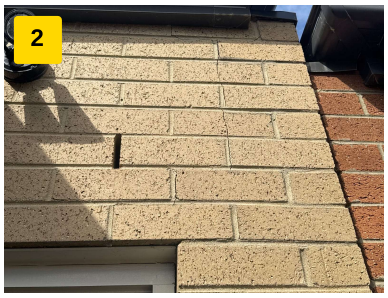
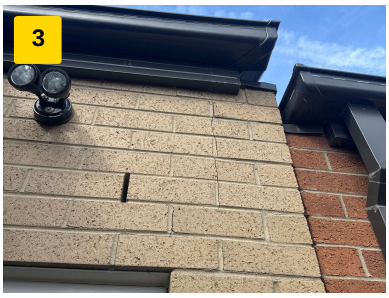
Scope of Works

Strength Engineering recommends the following:

- Refer to Observation Record 1
- The cracking does not pose a risk to the integrity of the structure and does not require repair. However, it can be filled with a suitable repair product such as low-viscosity epoxy resin.
- The low area is to be brought back up to the level of the surrounding floor using a concrete overlay.

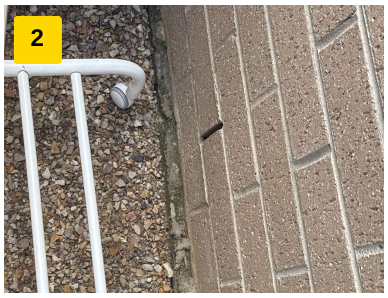
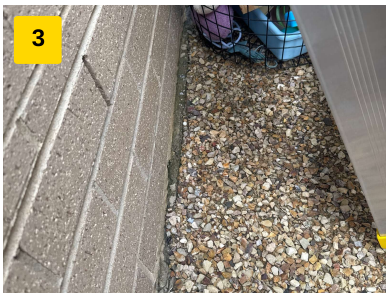
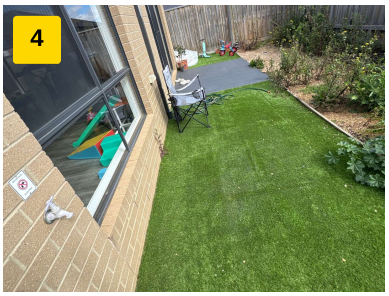
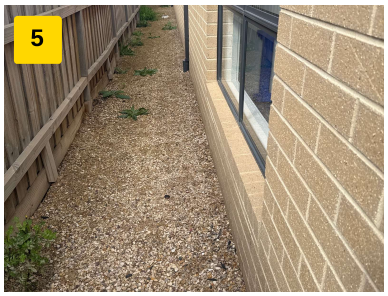
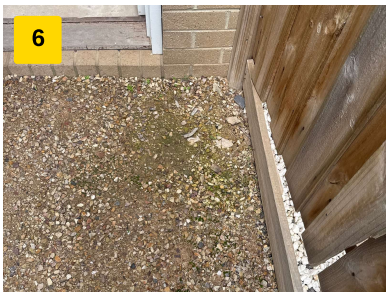
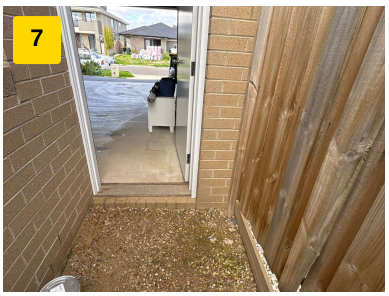
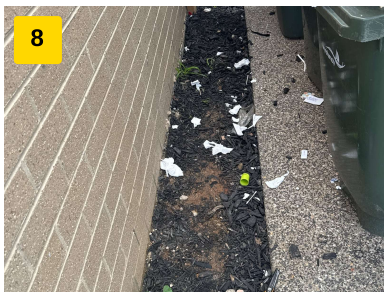
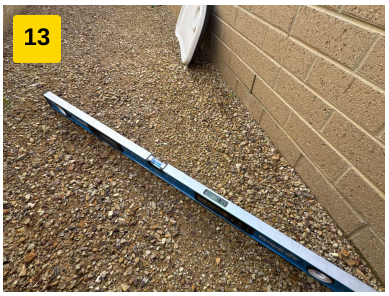
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3.2.5. Observation 5

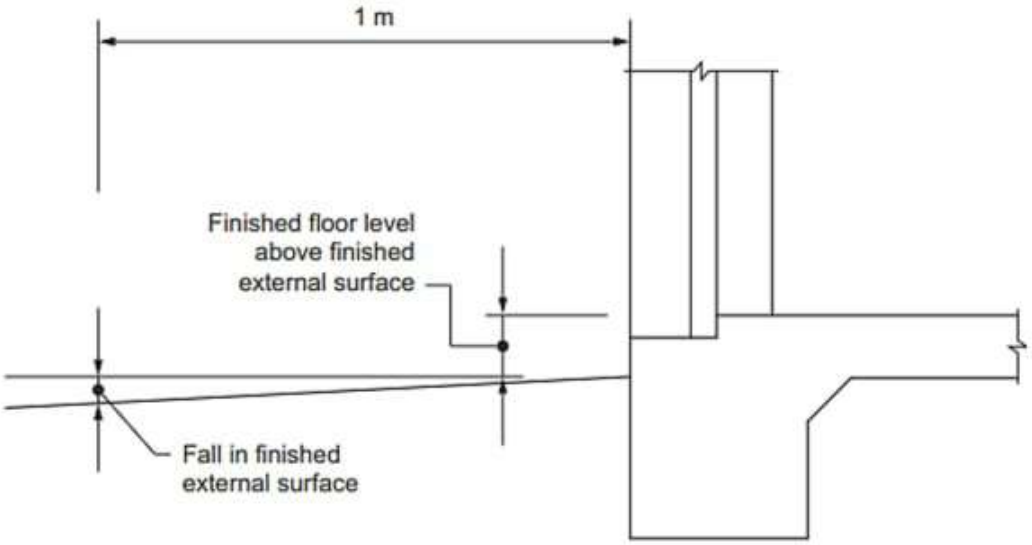
Observation No.	5
Photos	  
Observation & Comment	The following observation was made along the external areas of the subject property: • Minor stepped cracking transitioning into vertical cracking through the mortar joints with a width of up to approximately 1 mm, classified as damage category 1 (Refer to Table C1 in Appendix A), was observed to the external brick work above the garage entrance door.
Causation	Refer to Observation Record 1
Scope of Works	Strength Engineering recommends the following: • Refer to Observation Record 1 • A qualified masonry repair specialist is to be engaged to rectify the observed cracking by repointing the mortar joints or applying low-viscosity resin where appropriate.

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3.2.6. Observation 6

Observation No.	6
Photos	            
Observation & Comment	Signs of inadequate site surface drainage were observed around the perimeter of the subject property. Poor drainage can result in water ponding adjacent to the footings, increasing soil moisture levels and potentially contributing to differential ground movement. The subject property is not located in a low rainfall intensity area.

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	(a) Slab-on-ground — finished ground level adjacent to a building: the external finished surface surrounding the slab must be drained to move *surface water* away from the building and graded to give a slope of not less than (see Figure 3.3.3a) — (i) 25 mm over the first 1 m from the building— (A) in *low rainfall intensity areas* for surfaces that are reasonably impermeable (such as concrete or clay paving); or (B) for any reasonably impermeable surface that forms part of an access path or ramp provided for the purposes of Clauses 1.1(2) or (4)(c) of the ABCB Standard for Livable Housing Design; or (ii) 50 mm over the first 1 m from the building in any other case.
Causation	N/A
Scope of Works	Provide a certified civil and stormwater engineering design for adequate site surface drainage, prepared by Strength Engineering or another qualified structural engineer, and engage a suitably qualified concreter and plumber to implement the works in accordance with the certified design and the ABCB Housing Provisions Standard 2025 Cl. 3.3.3. The subgrade must be properly prepared, including appropriate compaction and removal of unsuitable material, to support the drainage works and minimise the risk of future movement. When implementing site surface drainage, care must be taken to ensure that it does not adversely affect neighbouring properties. **Figure 3.3.3a: Site surface drainage** 

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3.2.7. Observation 7

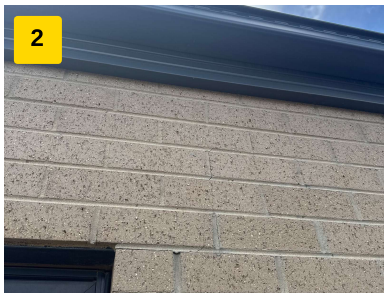
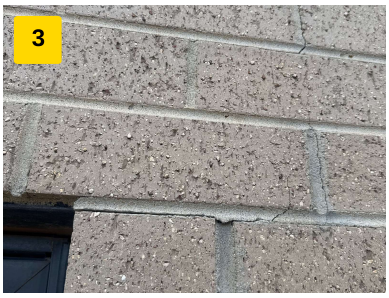
Observation No.	7
Photos	   
Observation & Comment	The following observation was made along the northern boundary of the subject property: • Separation/gap was visible between the concrete pavement and the external brick work.
Causation	This is likely due to one or more of the following: • Refer to Observation Record 1 • The gap between the concrete pavement and external brick work is due to subgrade movement beneath the external paving, likely caused by changes in soil moisture, reactive clay shrink–swell behaviors, or inadequate compaction of the base material.
Scope of Works	Strength Engineering recommends the following: • Refer to Observation Record 1

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	• Re-establish joint with abelflex or backing rod between paving and brick wall with flexible sealant to mitigate moisture ingress.

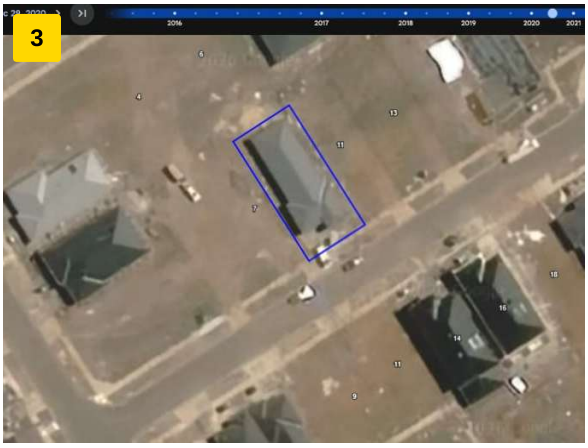
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3.2.8. Observation 8

Observation No.	8
Photos	    
Observation & Comment	The following observation was made along the western boundary external wall of the subject property: • Minor stepped cracking and vertical cracking, with a width of less than 1 mm, classified as damage category 1 (Refer to Table C1 in Appendix A), was observed along the western external boundary wall.
Causation	Refer to Observation Record 1
Scope of Works	Strength Engineering recommends the following: • Refer to Observation Record 1 • Hairline cracks do not currently appear to pose a structural risk and can be repaired in accordance with Observation Record 5

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3.2.9. Observation 9

Observation No.	9
Photos	   
Observation & Comment	Following receipt of the geotechnical investigation provide in the Engineering Drawings, which classifies the site as Class 'P' in accordance with AS 2870-2011, historical satellite imagery was reviewed to assist in understanding the nature of the underlying soil conditions and the basis for the classification. The imagery indicates that the site has been subject to filling prior to construction, which is consistent with the structural documentation providing for the waffle slab to be supported on bored piers founded through the fill and into the underlying natural material. Understanding the reactivity of the site is necessary in order to assess the observed movement, as fill material of this nature is highly susceptible to volumetric change with variation in moisture content. Accordingly, to limit further footing movement and to improve the long-term performance of the structure, moisture-related issues beneath and around the building are to be identified and eliminated.
Causation	N/A
Scope of Works	N/A

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4. Conclusion

Strength Engineering Pty Ltd has been engaged by the client to conduct an inspection at the subject site and provide our professional opinion as to the cause(s) of the observed damage. Records of the findings of the investigation have been provided in this report. Our professional recommendations are presented to aid the client in planning for rectification and or further investigations. The information provided in this report has addressed the deliverables to enable the client to deal with the matter appropriately. For inquiries related to this report please contact our office via contact@strengtheng.com.au.

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Notes & Disclaimer

This report is to be used as a set of instructions only and is not a certificate of compliance for any design, demolition, or construction works.

Visual limited*: This report is based on a visual inspection limited only to the areas and sections of the property fully accessible and visible to the inspector on the day of the inspection. The Inspection did not include breaking apart, dismantling, removing, or moving objects including, but not limited to foliage or personal possessions. The Inspector did not dig, gouge, force, or perform any other intrusive or destructive procedure. At the time of inspection, no information has been provided to indicate the specifications of any structural element on site. Visible structural elements cannot be destructively probed or hit without the written permission of the property owners. A qualified inspector has inspected and produced the report based on the prevailing conditions at the time of inspection. Please refer to this office if you have any inquiries regarding this report.

This report was completed in consideration of the requirements of THE BUILDING REGULATIONS 2018, NCC 2025 VOL 2 & AUSTRALIAN STANDARDS: AS/NZS 1170.1-2002, AS/NZS 1170.2-2021, AS 1684.2-2021, AS 2870-2011, AS 3600-2018, AS 3700-2018, AS 4100-2020, AS 4349.1-2007, AS 2858-2008, and AS 2082-2000.

This report was prepared within the limitations and conditions specified in Australian Standard AS 4349.1 - 2007 Pre-purchase Inspections - Residential buildings.

This report reflects the state of the structure at the time of inspection. This report does not outline any information that will guarantee that problems will not arise in the future. Different access, vicinity, and environmental conditions to that experienced at the time of inspection could reveal evidence of defects that may otherwise be undetectable.

This report is not an all-encompassing report dealing with the building from every aspect. It is a reasonable attempt to identify any obvious or significant defects apparent at the time of the inspection. Whether a defect is considered significant or not, depends to a large extent, upon the age and type of the building inspected.

This report contains a general description of the observations that Strength Engineering has deemed relevant to the scope of work provided by the client. It is not intended to be a complete list of building damage and/or defects.

This report has been prepared on behalf of and for the exclusive use of the Client, and is subject to and issued in accordance with the agreement between the Client and Strength Engineering Pty Ltd. Strength Engineering Pty Ltd accepts no liability or responsibility whatsoever for any use of or reliance upon this report by any third party. Any copying of this report to external parties requires the permission of the Client and Strength Engineering Pty Ltd.

This report is not a certificate of compliance of the property under any act, regulation, ordinance, local law, or bylaw. It is not a warranty against problems developing with the building in the future. This report does not include the detection and identification of unauthorised or illegal building, plumbing, or electrical work, or work not compliant with building regulations. Concerning minor defects, the inspection is limited to reporting on their overall extent not listing each one.

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Appendice A: Classification of Damage due to Foundation Movements

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TABLE C1

CLASSIFICATION OF DAMAGE WITH REFERENCE TO WALLS

Description of typical damage and required repair	Approximate crack width limit (see Note 1)	Damage category
Hairline cracks	<0.1 mm	0 Negligible
Fine cracks that do not need repair	<1 mm	1 Very slight
Cracks noticeable but easily filled. Doors and windows stick slightly	<5 mm	2 Slight
Cracks can be repaired and possibly a small amount of wall will need to be replaced. Doors and windows stick. Service pipes can fracture. Weather tightness often impaired	5 mm to 15 mm (or a number of cracks 3 mm or more in one group)	3 Moderate
Extensive repair work involving breaking out and replacing sections of walls, especially over doors and windows. Window frames and door frames distort. Walls lean or bulge noticeably, some loss of bearing in beams. Service pipes disrupted	15 mm to 25 mm but also depends on number of cracks	4 Severe

NOTES:

1 Where the cracking occurs in easily repaired plasterboard or similar clad-framed partitions, the crack width limits may be increased by 50% for each damage category.

2 Crack width is the main factor by which damage to walls is categorized. The width may be supplemented by other factors, including serviceability, in assessing category of damage.

3 In assessing the degree of damage, account shall be taken of the location in the building or structure where it occurs, and also of the function of the building or structure.

TABLE C2

CLASSIFICATION OF DAMAGE WITH REFERENCE TO CONCRETE FLOORS

Description of typical damage	Approx. crack width limit in floor	Change in offset from a 3 m straightedge centred over defect (see Note 1)	Damage category
Hairline cracks, insignificant movement of slab from level	<0.3 mm	<8 mm	0 Negligible
Fine but noticeable cracks. Slab reasonably level	<1.0 mm	<10 mm	1 Very slight
Distinct cracks. Slab noticeably curved or changed in level	<2.0 mm	<15 mm	2 Slight
Wide cracks. Obvious curvature or change in level	2 mm to 4 mm	15 mm to 25 mm	3 Moderate
Gaps in slab. Disturbing curvature or change in level	4 mm to 10 mm	>25 mm	4 Severe

NOTES:

1 The straightedge is centred over the defect, usually, and supported at its ends by equal height spacers. The change in offset is then measured relative to this straightedge, which is not necessarily horizontal.

2 Local deviation of slope, from the horizontal or vertical, of more than 1:100 will normally be clearly visible. Overall deviations in excess of 1:150 is undesirable.

3 Account should be taken of the past history of damage in order to assess whether it is stable or likely to increase.

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Appendice B: CSIRO Guide

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Foundation Maintenance and Footing Performance:

A Homeowner's Guide

CSIRO

BTF 18
replaces
Information
Sheet 10/91

Buildings can and often do move. This movement can be up, down, lateral or rotational. The fundamental cause of movement in buildings can usually be related to one or more problems in the foundation soil. It is important for the homeowner to identify the soil type in order to ascertain the measures that should be put in place in order to ensure that problems in the foundation soil can be prevented, thus protecting against building movement.

This Building Technology File is designed to identify causes of soil-related building movement, and to suggest methods of prevention of resultant cracking in buildings.

Soil Types

The types of soils usually present under the topsoil in land zoned for residential buildings can be split into two approximate groups – granular and clay. Quite often, foundation soil is a mixture of both types. The general problems associated with soils having granular content are usually caused by erosion. Clay soils are subject to saturation and swell/shrink problems.

Classifications for a given area can generally be obtained by application to the local authority, but these are sometimes unreliable and if there is doubt, a geotechnical report should be commissioned. As most buildings suffering movement problems are founded on clay soils, there is an emphasis on classification of soils according to the amount of swell and shrinkage they experience with variations of water content. The table below is Table 2.1 from AS 2870, the Residential Slab and Footing Code.

Causes of Movement

Settlement due to construction
There are two types of settlement that occur as a result of construction:

- Immediate settlement occurs when a building is first placed on its foundation soil, as a result of compaction of the soil under the weight of the structure. The cohesive quality of clay soil mitigates against this, but granular (particularly sandy) soil is susceptible.
- Consolidation settlement is a feature of clay soil and may take place because of the expulsion of moisture from the soil or because of the soil's lack of resistance to local compressive or shear stresses. This will usually take place during the first few months after construction, but has been known to take many years in exceptional cases.

These problems are the province of the builder and should be taken into consideration as part of the preparation of the site for construction. Building Technology File 19 (BTF 19) deals with these problems.

Erosion
All soils are prone to erosion, but sandy soil is particularly susceptible to being washed away. Even clay with a sand component of say 10% or more can suffer from erosion.

Saturation
This is particularly a problem in clay soils. Saturation creates a bog-like suspension of the soil that causes it to lose virtually all of its bearing capacity. To a lesser degree, sand is affected by saturation because saturated sand may undergo a reduction in volume – particularly imported sand fill for bedding and blinding layers. However, this usually occurs as immediate settlement and should normally be the province of the builder.

Seasonal swelling and shrinkage of soil
All clays react to the presence of water by slowly absorbing it, making the soil increase in volume (see table below). The degree of increase varies considerably between different clays, as does the degree of decrease during the subsequent drying out caused by fair weather periods. Because of the low absorption and expulsion rate, this phenomenon will not usually be noticeable unless there are prolonged rainy or dry periods, usually of weeks or months, depending on the land and soil characteristics.

The swelling of soil creates an upward force on the footings of the building, and shrinkage creates subsidence that takes away the support needed by the footing to retain equilibrium.

Shear failure
This phenomenon occurs when the foundation soil does not have sufficient strength to support the weight of the footing. There are two major post-construction causes:

- Significant load increase.
- Reduction of lateral support of the soil under the footing due to erosion or excavation.
- In clay soil, shear failure can be caused by saturation of the soil adjacent to or under the footing.

GENERAL DEFINITIONS OF SITE CLASSES

Class	Foundation
A	Most sand and rock sites with little or no ground movement from moisture changes
S	Slightly reactive clay sites with only slight ground movement from moisture changes
M	Moderately reactive clay or silt sites, which can experience moderate ground movement from moisture changes
H	Highly reactive clay sites, which can experience high ground movement from moisture changes
E	Extremely reactive sites, which can experience extreme ground movement from moisture changes
A to P	Filled sites
P	Sites which include soft soils, such as soft clay or silt or loose sands; landslip; mine subsidence; collapsing soils; soils subject to erosion; reactive sites subject to abnormal moisture conditions or sites which cannot be classified otherwise

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Tree root growth
Trees and shrubs that are allowed to grow in the vicinity of footings can cause foundation soil movement in two ways:

- Roots that grow under footings may increase in cross-sectional size, exerting upward pressure on footings.
- Roots in the vicinity of footings will absorb much of the moisture in the foundation soil, causing shrinkage or subsidence.

Unevenness of Movement

The types of ground movement described above usually occur unevenly throughout the building's foundation soil. Settlement due to construction tends to be uneven because of:

- Differing compaction of foundation soil prior to construction.
- Differing moisture content of foundation soil prior to construction.

Movement due to non-construction causes is usually more uneven still. Erosion can undermine a footing that traverses the flow or can create the conditions for shear failure by eroding soil adjacent to a footing that runs in the same direction as the flow.

Saturation of clay foundation soil may occur where subfloor walls create a dam that makes water pond. It can also occur wherever there is a source of water near footings in clay soil. This leads to a severe reduction in the strength of the soil which may create local shear failure.

Seasonal swelling and shrinkage of clay soil affects the perimeter of the building first, then gradually spreads to the interior. The swelling process will usually begin at the uphill extreme of the building, or on the weather side where the land is flat. Swelling gradually reaches the interior soil as absorption continues. Shrinkage usually begins where the sun's heat is greatest.

Effects of Uneven Soil Movement on Structures

Erosion and saturation
Erosion removes the support from under footings, tending to create subsidence of the part of the structure under which it occurs. Brickwork walls will resist the stress created by this removal of support by bridging the gap or cantilevering until the bricks or the mortar bedding fail. Older masonry has little resistance. Evidence of failure varies according to circumstances and symptoms may include:

- Step cracking in the mortar beds in the body of the wall or above/below openings such as doors or windows.
- Vertical cracking in the bricks (usually but not necessarily in line with the vertical beds or perpends).

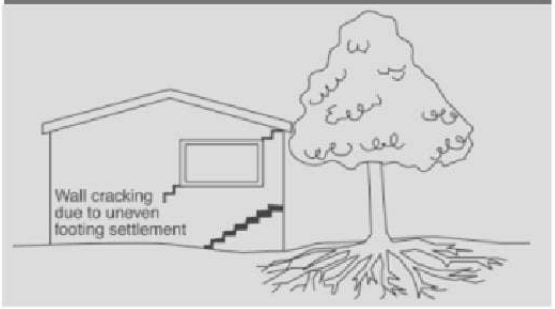
Isolated piers affected by erosion or saturation of foundations will eventually lose contact with the bearers they support and may tilt or fall over. The floors that have lost this support will become bouncy, sometimes rattling ornaments etc.

Seasonal swelling/shrinkage in clay
Swelling foundation soil due to rainy periods first lifts the most exposed extremities of the footing system, then the remainder of the perimeter footings while gradually permeating inside the building footprint to lift internal footings. This swelling first tends to create a dish effect, because the external footings are pushed higher than the internal ones.

The first noticeable symptom may be that the floor appears slightly dished. This is often accompanied by some doors binding on the floor or the door head, together with some cracking of cornice mitres. In buildings with timber flooring supported by bearers and joists, the floor can be bouncy. Externally there may be visible dishing of the hip or ridge lines.

As the moisture absorption process completes its journey to the innermost areas of the building, the internal footings will rise. If the spread of moisture is roughly even, it may be that the symptoms will temporarily disappear, but it is more likely that swelling will be uneven, creating a difference rather than a disappearance in symptoms. In buildings with timber flooring supported by bearers and joists, the isolated piers will rise more easily than the strip footings or piers under walls, creating noticeable doming of flooring.

Trees can cause shrinkage and damage



As the weather pattern changes and the soil begins to dry out, the external footings will be first affected, beginning with the locations where the sun's effect is strongest. This has the effect of lowering the external footings. The doming is accentuated and cracking reduces or disappears where it occurred because of dishing, but other cracks open up. The roof lines may become convex.

Doming and dishing are also affected by weather in other ways. In areas where warm, wet summers and cooler dry winters prevail, water migration tends to be toward the interior and doming will be accentuated, whereas where summers are dry and winters are cold and wet, migration tends to be toward the exterior and the underlying propensity is toward dishing.

Movement caused by tree roots
In general, growing roots will exert an upward pressure on footings, whereas soil subject to drying because of tree or shrub roots will tend to remove support from under footings by inducing shrinkage.

Complications caused by the structure itself
Most forces that the soil causes to be exerted on structures are vertical – i.e. either up or down. However, because these forces are seldom spread evenly around the footings, and because the building resists uneven movement because of its rigidity, forces are exerted from one part of the building to another. The net result of all these forces is usually rotational. This resultant force often complicates the diagnosis because the visible symptoms do not simply reflect the original cause. A common symptom is binding of doors on the vertical member of the frame.

Effects on full masonry structures
Brickwork will resist cracking where it can. It will attempt to span areas that lose support because of subsided foundations or raised points. It is therefore usual to see cracking at weak points, such as openings for windows or doors.

In the event of construction settlement, cracking will usually remain unchanged after the process of settlement has ceased.

With local shear or erosion, cracking will usually continue to develop until the original cause has been remedied, or until the subsidence has completely neutralised the affected portion of footing and the structure has stabilised on other footings that remain effective.

In the case of swell/shrink effects, the brickwork will in some cases return to its original position after completion of a cycle, however it is more likely that the rotational effect will not be exactly reversed, and it is also usual that brickwork will settle in its new position and will resist the forces trying to return it to its original position. This means that in a case where swelling takes place after construction and cracking occurs, the cracking is likely to at least partly remain after the shrink segment of the cycle is complete. Thus, each time the cycle is repeated, the likelihood is that the cracking will become wider until the sections of brickwork become virtually independent.

With repeated cycles, once the cracking is established, if there is no other complication, it is normal for the incidence of cracking to stabilise, as the building has the articulation it needs to cope with the problem. This is by no means always the case, however, and monitoring of cracks in walls and floors should always be treated seriously.

Upheaval caused by growth of tree roots under footings is not a simple vertical shear stress. There is a tendency for the root to also exert lateral forces that attempt to separate sections of brickwork after initial cracking has occurred.

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Appendice C: Floor Level Survey

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Appendice D: Structural Engineering Document

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STANDARD SPECIFICATION NOTES: PROPOSED NEW RESIDENCE

GENERAL NOTES:

G1. This drawing shall be read in conjunction with the architectural plans and specifications.
G2. The builder is to check and be responsible for the correctness of all dimensions and any discrepancy is to be reported. Do not scale drawings.
G3. Stability of the building during construction and all excavations in the vicinity of neighbouring buildings is the sole responsibility of the builder.
G4. All workmanship and materials are to be in accordance with current S.A. codes BCA and local council requirements.
G5. Design Liveloads are 1.5 kPA (DOMESTIC)
G6. Foundation material to be approved before pouring concrete for safe bearing capacity of 150 KPa. minimum.

CONCRETE NOTES:

C1. Concrete quality as per AS 3600 shall be as follows, re. 28 days type testing F'c 20 MPa Maximum slump 75 mm and maximum aggregate size of 20 mm.
C2. Concrete cover to all reinforcement (finishes not included)

ELEMENT	SHELTERED	EXPOSED	NO-FORM
slab & wall	20 mm	30 mm	65 mm
beams	25 mm	40 mm	65 mm
columns	40 mm	50 mm	75 mm
footings	—	65 mm	75 mm

C3. Bar and Mesh designations:
ø – Structural grade round bar to AS 1302
N. – Structural grade deformed bar to AS 1302
H. – Hard grade deformed bar to AS 1302
CW. – Cold Worked deformed bar to AS 1302
F. – Fabric to AS 1303 and AS 1304
C4. All concrete shall be mechanically vibrated and the vibrator shall not be used to vibrate the forms nor shall it be used to spread concrete.
C5. Depth of beam is given first and includes slab thickness not including any finishes that may be applied.
C6. Splices in reinforement shall be sufficient to develop the full strength of the reinforement without displacement from structural location. Laps to fabric shall be two transverse wires plus 100 mm.
C7. Reinforement shall be accurately and firmly fixed in position shown to give support and cover specified during all operations of pouring, etc.
C8. Conduits, pipes, etc., must not be placed in concrete cover and no holes other than those shown on the drawings shall be permitted.
C9. Formwork shall remain in position for a minimum of 14 days and where slabs and beams are to support brickwork over, formwork and props must be removed before commencement of this brickwork.
C10. Concrete must be cured for 7 days using water ponding wet sand or otherwise approved methods of adequate curing unless noted otherwise.
C11. Wheelbarrow or pump pipes must be supported directly from the formwork.
C12. Additives must not be be added to concrete without the engineer's approval.
C13. Field welding of reinforcement is only permitted where shown on the drawings or otherwise approved.
C14. Load–bearing brickwork shall be seperated from concrete by the use of Malthoid or similar.

TIMBER NOTES:

T1. All timber shall be dry, ie. less than 15% moisture content at the time of construction and shall be protected and/or treated as noted.
T2. All timber used shall comply with AS 1720, 1684 and shall have a minimum stress grade MGP10 .
T3. All timber fixing and bracing as per AS 1684.

BRICK NOTES:

B1. All brickwork shall be constructed in accordance with AS 3700.
B2. All load bearing bricks shall have a minimum crushing strength of 50 MPa.
B3. No chases shall be cut into load bearing brickwork without approval of consulting engineer.
B4. All brickwork shall be constructed in motar composing 1 part cement 1 part lime 6 part sand.
B5. Where walls are non–load bearing at either horizontal or vertical faces they shall be seperated from concrete by 10 mm thick conxite.
B6. All brickwork shall be articulated at 6m centres(min).
B7. All rendered brickwork shall distinguish the articulation joints. Do not render/bag over articulation joints.

STEEL NOTES:

S1. All steelwork shall be in accordance with the current code.
S2. Welds shall be 6 mm cont. fillet or single vee butt except at toes of rolled steel sections where they shall be the max. size permitted by the welding code. All butt welds to develop the full strength of min. member connected.
S3. Bolts shall be mild steel in 1.5 mm clearance holes. Where high strength (HS) bolts are specified they shall be in accordance with AS 1252 and tightened by an approved method.
S4. Steelwork shall be given one shop coat of primer except that none shall be applied at contact surfaces where HS bolts are used.
S5. The fabricator shall submit shop detail drawings to the engineer for approval of connections before commencing fabrication. These shall be in accordance with the typical details shown.
S6. The fabricator shall provide all cleats and holes for the connection of purlings, girts, brick ties, etc. Braces and ties shall have true intersection.
S7. Ties and braces shall be connected to 10 mm gusset plates with 2–12 mm ø bolts each end.
S8. Steelwork below floor level shall be encased 75 mm minimum in concrete.
S9. Columns shall be bedded on 1:1 cement grout after plumbing and levelling on steel packers.
S10. Members bent during fabrication, transport or handling will not be accepted.
S11. Provide sag rods and/or struts to manufacturer's rec. to purlins and girts.
S12. Steelwork where concrete encased shall be wrapped with SL62 mesh and have 50 mm cover.
S13. Steelwork to be protected of exposed areas as per BCA, clause 3.4.4.2
Refer to BCA/NCC for corrosion specifications.
S14. Steel lintels to have a minimum end bearing of 110mm, unless noted otherwise.

PREPARATION OF SUB-BASE FOR SLABS ON GROUND

P1. All preparation of sub–base for slabs on ground to me in accordance AS 2870.
P2. Clear area under slab of all top soil containing humus and vegetable matter 100mm min.
P3. Provide fill under slab where required to produce finished levels as shown on plans. All fill shall be imported and conform to R.C.A. standard specification for class 3 crushed rock (20mm nom. size). Fill to be compacted in 200mm max layers to 95% of the modified max dry density (M.M.D.D.) when tested in accordance with AS 1289.
P4. The upper layer of the cut surface shall be within 85% to 115% of optimum moisture content and to be properly compacted to 95% M.M.D.D.
P5. A 50mm min. base course of packing sand shall be spread over the sub–base and to be thoroughly rolled and compacted to a smooth level surface. The sand shall be moistened prior to the placement of a 0.2mm polythene membrane in 3600mm min. wide sheets lapped 150mm and jointed with 75mm wide pressure sensitive tape. The tape shall be laid under all slabs and walls in contact with the ground.
P6. The total fill beneath the slab panels must not exceed 400mm, ie. the sum of existing fill plus any new filling placed together must not exceed 400mm max. Fill to be located as per clause 6.4.2 AS 2870.

INSPECTIONS:

No responsiblity shall be taken unless the work is inspected and approved during construction; all inspection required shall be comfirmed 24 hours in advance of time required.

SITE DRAINAGE

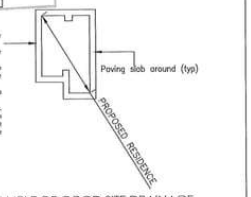
Sites should be drained so that water cannot pond against or near the building. The ground immediately adjacent to the building should be graded to fall 50mm over the first metre. Where this is impracticable (ie: on several sloping sites) use A.G. drains adjacent to footings, where the ground falls towards the building.(refer to details below). Good drainage and paving around house is recommended to ensure long term performance of residence.



OWNER'S RESPONSIBILITY

IT IS THE OWNER'S RESPONSIBILITY:

(i) Not to plant or allow the growth of trees or dense shrubs too close to the footings.
(ii) Not to excessively water the soil near the footings or allow it to dry excessively.
(iii) To maintain good site drainage and not to allow plumbing leaks or water ponding around the building.
(iv) To follow all post construction recommendations.
(v) Not to pave, alter any part of the structure, carry out any excavations or landscaping which alters the site drainage (resulting in water not being diverted away from footings) and the foundation conditions.



EXAMPLE OF GOOD SITE DRAINAGE

No.	Subject	Date	Initial
B1	SUBMISSION FOR BUILDING PERMIT	21.08.20	

REVISIONS

TITLE GENERAL NOTES

LOT 114 FURNESS CLOSE
WYNDHAM VALE

Design	A.SO	Date	21.08.20
Drawn	R.F	Scale	N/A
		Drawing No.	20041
Sheet No. 1 of 8		Rev.	B1

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NOTE: ENSURE WORKING PLATFORM / FILLING MATERIAL IS WELL COMPACTED PRIOR TO PLACEMENT OF WAFFLE PODS. ANY FILLING PLACED DUE TO CUT/FILL OPERATION SHALL BE COMPACTED IN 150mm MAX. LAYERS BY REPEATED ROLLING BY AN EXCAVATOR IN ORTHOGONAL DIRECTIONS. DEPTH OF FILL SHALL NOT EXCEED 300mm.

NOTE: REFER ARCHITECTURAL DRAWINGS FOR ARTICULATION JOINT LOCATIONS AT 6000mm MAX CENTRES.

REFER TO THE ARCHITECT'S DRAWINGS FOR FINISHED FLOOR LEVELS AND SIZE AND EXACT LOCATIONS OF STEPS AND SETDOWNS.

NOTE:– ALL PLUMBING & SERVICE CONDUITS, PENETRATING SLAB TO BE FULLY LAGGED SO AS TO ALLOW FOR DIFFERENTIAL MOVEMENT BETWEEN SLAB & PIPES

NOTE ALL MASONRY WALLS TO BE FULLY ARTICULATED IN STRICT ACCORDANCE WITH THE "CEMENT & CONCRETE ASSOCIATION NOTE C90 AND OR TECHNICAL NOTE 61."

NOTE : EXTERNAL & INTERNAL RIBS WIDER THAN 300mm SHALL BE REINFORCED WITH AN ADDITIONAL N16 BAR TOP & BOTTOM FOR EVERY ADDITIONAL 110mm IN WIDTH. (TYP.)

NOTE: DRAINAGE PROVISIONS CRITICAL

REFER PART 5.6.3 PART A TO D FOR DRAINAGE REQUIREMENTS THE BASE OF TRENCHES SHALL BE SLOPED WAY FROM BUILDING. TRENCHES SHALL BE BACKFILLED WITH CLAY IN THE TOP 300mm WITHIN 1.5m OF BUILDING, WHERE PIPES PASS UNDER FOOTINGS THE TRENCH SHALL BE BACKFILLED FULL DEPTH WITH CLAY TO ACT AS BARRIER FOR INGRESS OF WATER BENEATH FOOTING.



TYPICAL STRIP FOOTING TO SLAB CONNECTION DETAIL

THIS SITE SOIL CLASSIFICATION IS:

CLASS 'P'

IN ACCORDANCE WITH AS.2870 1996, AND SITE INVESTIGATIONS BY: ARC SOIL TESTING REFERENCE No. SR 2020–252 DATED: 28.07.20

THIS PLAN SHOULD BE READ IN CONJUNCTION WITH THIS REPORT.

ALL FOOTINGS ARE TO BE FOUNDED A MINIMUM OF 150 mm INTO THE STIFF CLAYS AS NOTED IN THE BORELOGS.

WARNING

BEWARE OF UNDERGROUND SERVICES
THE BUILDER IS TO DETERMINE EXACT LOCATIONS OF EXISTING UNDERGROUND SERVICES PRIOR ANY CONSTRUCTION ON SITE.

WHEN LOOSE FILL UNDER WAFFLE EXCEEDS 300mm

LOCATE 450mm DIAMETER BORED PIERS & 3600 CTRS & CORNERS & FOUNDED 500mm (MIN.) INTO STIFF CLAY MATERIAL (REFER TYPICAL DETAIL) AS NOTED IN THE BORELOGS.

SF1 FOOTINGS WHERE REQUIRED

ADOPT 300–470mm WIDE x 1200mm DEEP STRIP FOOTING WITH MIN 3–L16 TM

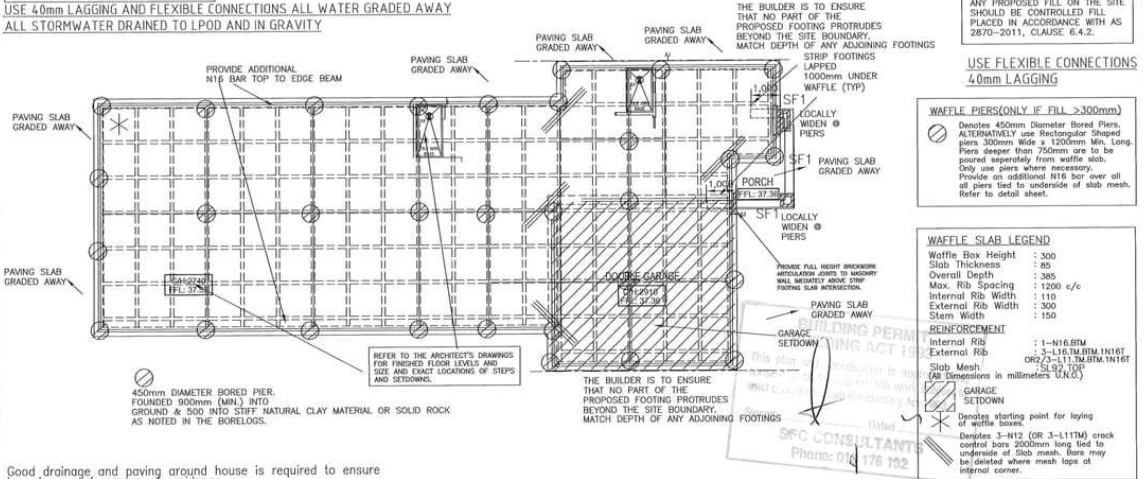
OR 500 WIDE X 900 DEEP ONTO SOLID ROCK FOR REFUSAL

NOTE:

ANY PROPOSED FILL ON THE SITE SHOULD BE CONTROLLED FILL PLACED IN ACCORDANCE WITH AS 2870–2011, CLAUSE 6.4.2.

USE 40mm LAGGING AND FLEXIBLE CONNECTIONS ALL WATER GRADED AWAY
ALL STORMWATER DRAINED TO LPOD AND IN GRAVITY

USE FLEXIBLE CONNECTIONS 40mm LAGGING



WAFFLE PIERS(ONLY IF FILL >300mm)

Denotes 450mm Diameter Bored Piers. ALTERNATIVELY use Rectangular Shaped piers 300mm Wide x 1200mm Min. Long. Piers deeper than 750mm are to be poured seperately from waffle slab. Only use piers where necessary. Provide an additional N16 bar over all all piers tied to underside of slab mesh. Refer to detail sheet.

WAFFLE SLAB LEGEND

Waffle Box Height	: 300
Slab Thickness	: 85
Overall Depth	: 385
Max. Rib Spacing	: 1200 c/c
Internal Rib Width	: 110
External Rib Width	: 300
Stem Width	: 150

REINFORCEMENT

Internal Rib	: 1–N16.BTM
External Rib	: 3–L16.TM.BTM.1N16T OR2/3–L11.TM.BTM.1N16T
Slab Mesh	: SL92 TOP

(All Dimensions in millimeters U.N.O.)

GARAGE SETDOWN

Denotes starting point for laying of waffle boxes.

Denotes 3–N12 (OR 3–L11TM) crack control bars 2000mm long tied to underside of Slab mesh. Bars may be deleted where mesh laps at internal corner.

Good drainage and paving around house is required to ensure long term performance of residence.
OWNER MUST BE AWARE HOME MAINTENANCE REQUIREMENTS REFER CSIRO *& HEDRA FACTS SHEETS

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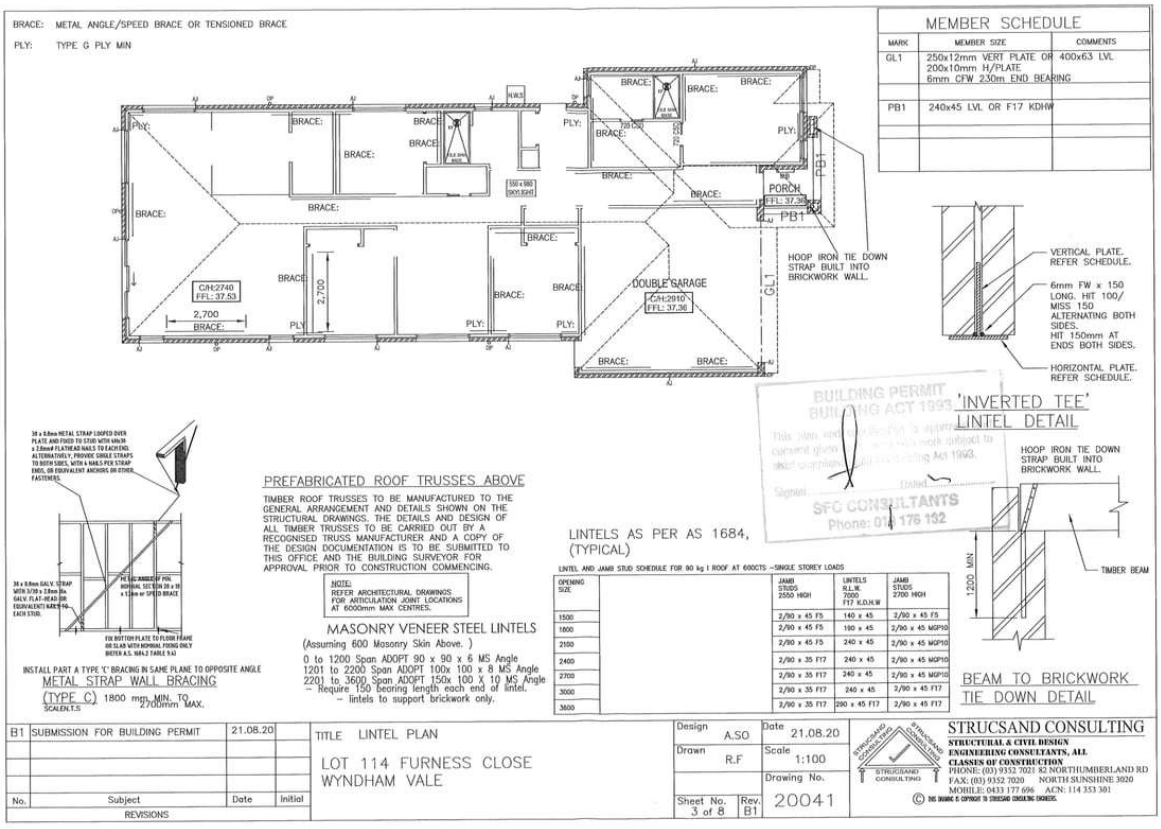
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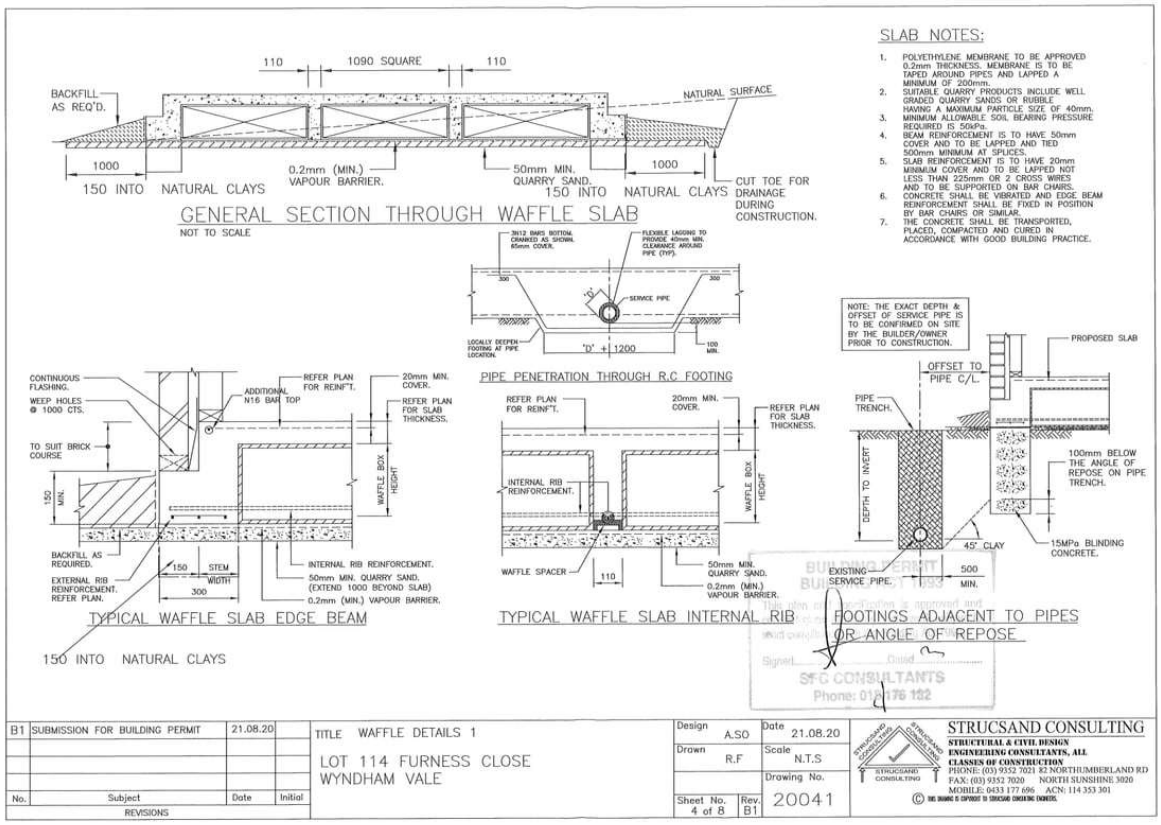
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WYNDHAM VALE

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		Drawing No.	20041
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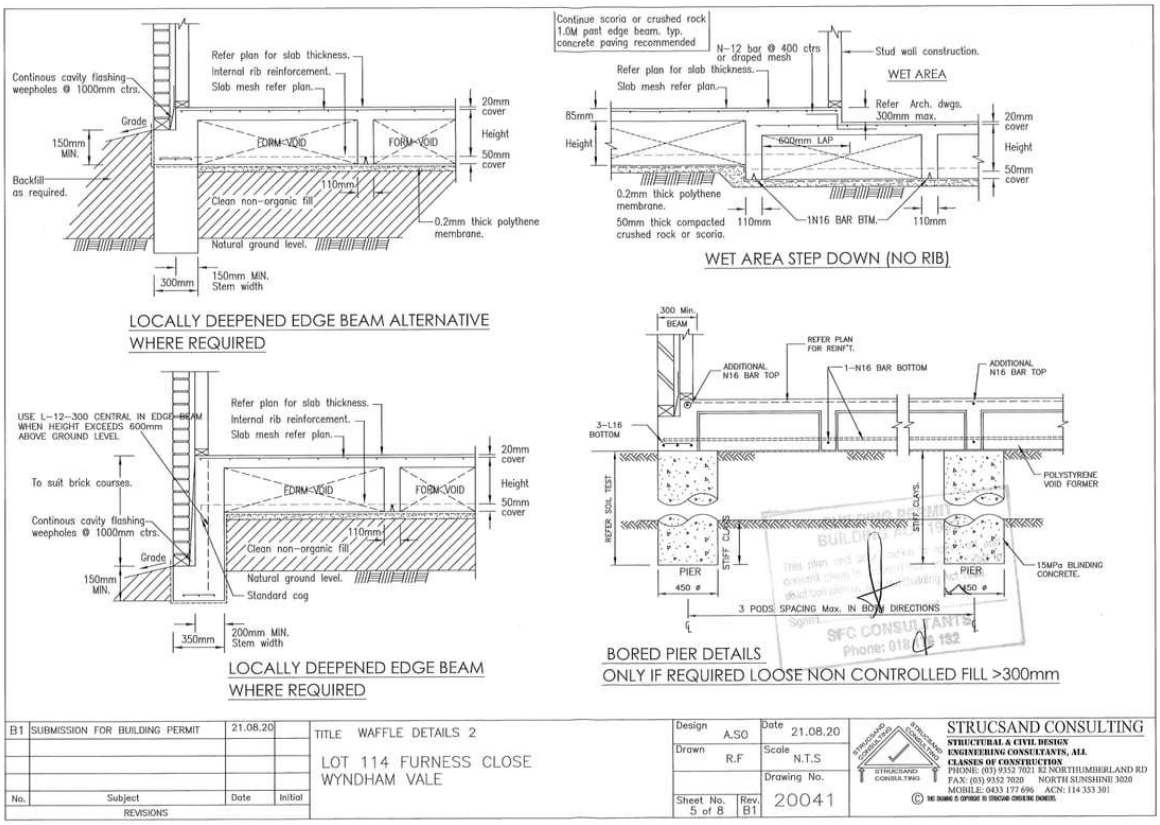
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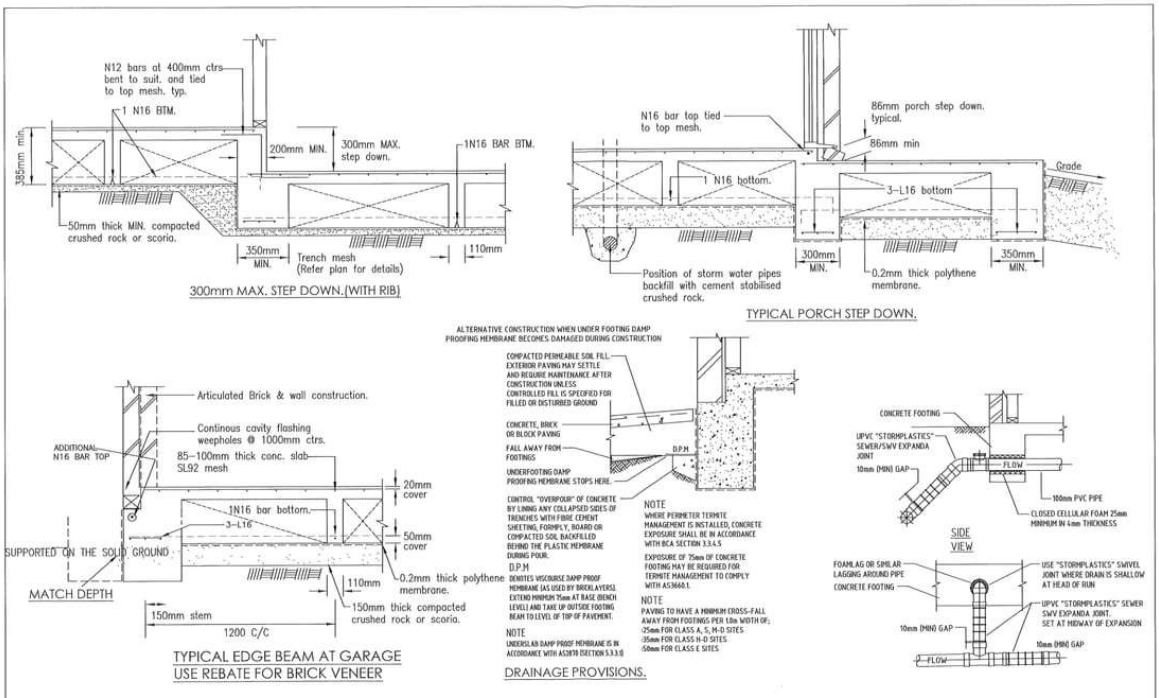




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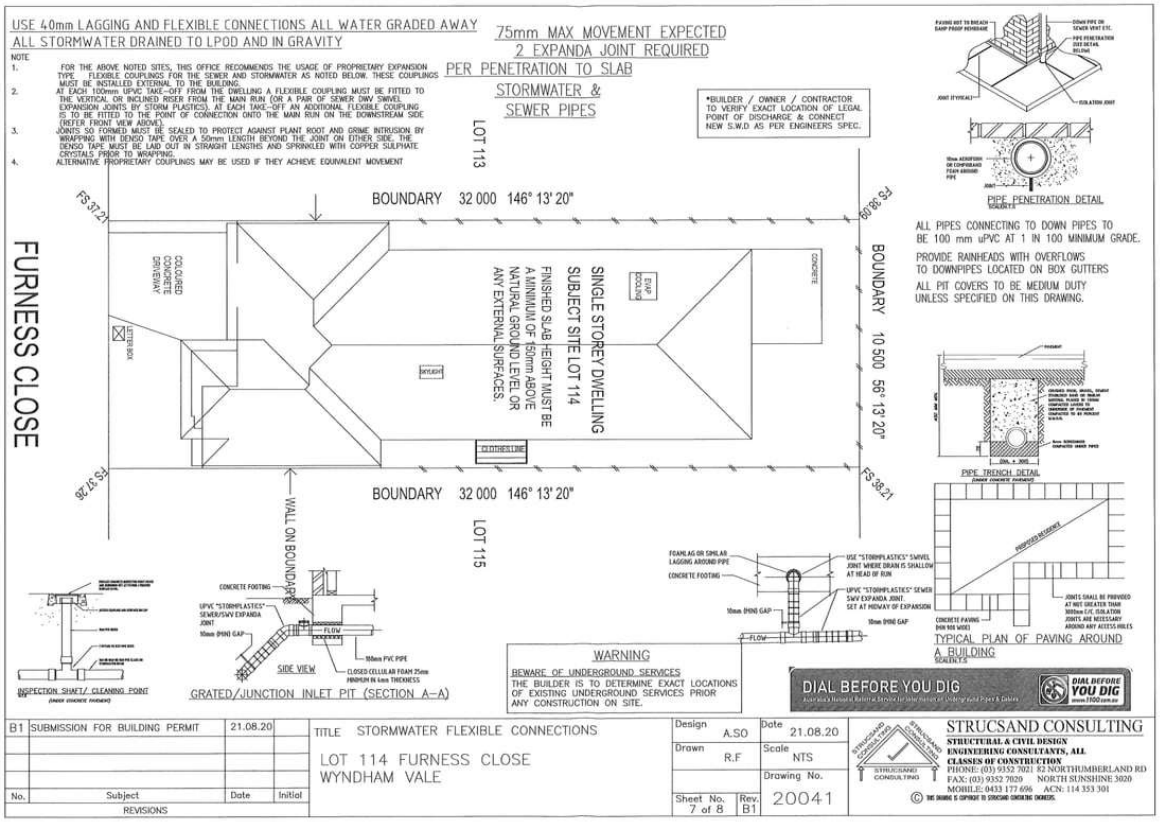


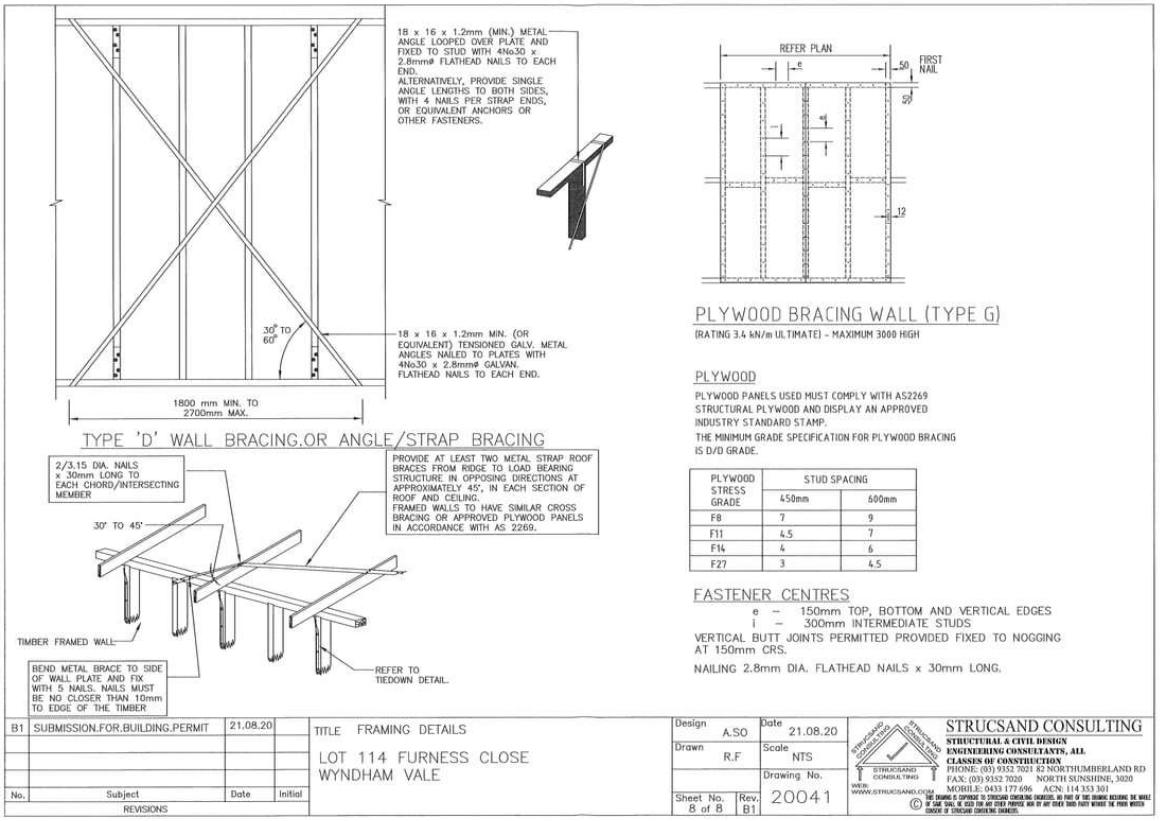
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Appendice E: Effects of Moisture Content on the Strength & Compression Properties of Clayey Soils

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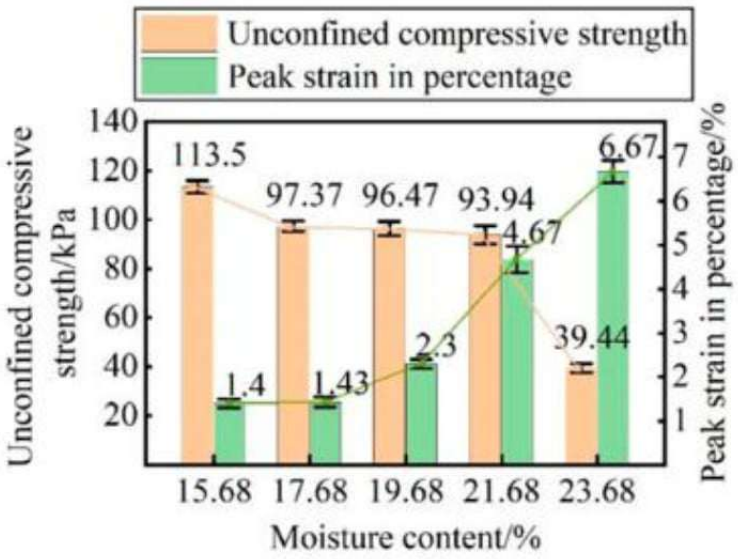


Figure 1: Curves of moisture content vs. strength and peak strain.

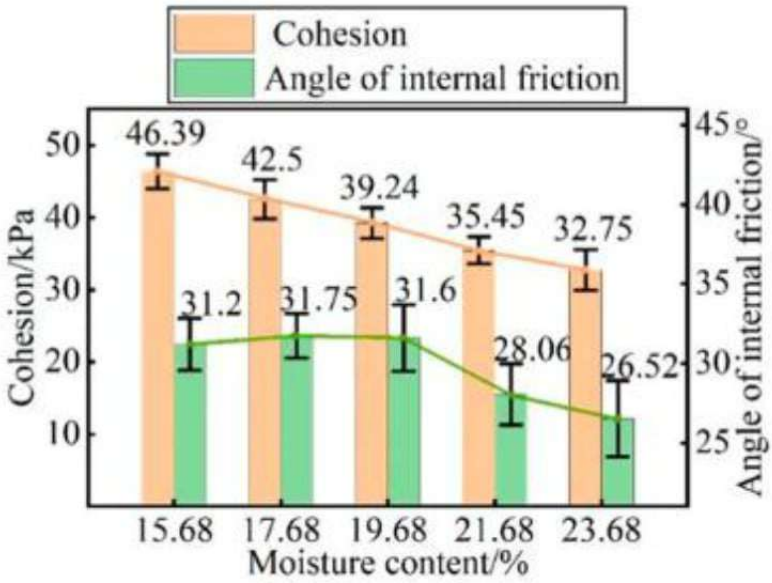


Figure 2: Curves of moisture content vs. cohesion and angle of internal friction.

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Appendice F: Building Defect Remediation - What is next Brochure V1.0?

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Forensic Investigation

Completed

You now know what's wrong and why it happened.
A detailed forensic engineering report outlining the observed distress, its causation, and initial scope of works recommendations has been delivered

Rectification Design

Request Quote

This phase will tell you exactly what to do and how to fix it. Custom engineering drawings and a Scope of Works document will provide a plan for rectifying the defect identified in Phase 1.

Construction Works

Request Quote

A registered builder will now carry out the remedial works as per the engineering plans, ensuring all works are done to standards. Upon completion, the defects will have been addressed and your structure is now physically repaired and compliant.

Maintenance & Monitoring (Optional)

You preserve asset value and prevent future issues. Periodic maintenance inspections and preventative repairs will help maintain the integrity of your building over time.